

Asia Pacific Equity Derivatives Strategy

KOSPI2 Gamma Reset, Not Cleared; Volatility Sticky, Less Explosive; Monetize Rich Volatility With KO Calls And Iron Condors

- Korea Short Gamma Resets, But Remains Meaningful**: The July deleveraging has taken the heat out of Korea's short-gamma squeeze, but not the risk. **Dealer short gamma is well below the YTD peak** of roughly US\$1.6bn, yet at **around US\$440mn remains well above the YTD low** of roughly US\$206mn and **still ETF-led**. A more subtle shift is emerging in composition: **Korea options are moving toward dealer neutral-to-long gamma**, especially in onshore and offshore **SK hynix options**, likely reflecting investors monetizing elevated volatility. The addition of ADR options gives investors another offshore hedging channel, while recent option selling is helping bring some volatility supply back. Lower SK hynix implied funding also confirms that leveraged demand has eased. The **options offset is still small versus ETF-driven short gamma**, and the spot/gamma profile suggests the risk of an ETF-led gamma rebuild remains if spot rallies.
- Korea Volatility: Less Explosive, Still Sticky**: The leverage unwind has **reduced the risk of another parabolic spot-up / volatility-up squeeze, but normalization is likely to take time**. Korea's volatility supply and demand imbalance is improving, not fixed: leveraged ETF AUM has reset, while structured-product issuance and early investor option selling are only starting to rebuild supply. Unlike Hong Kong or the US, **Korea has less volatility supply to force quick mean reversion, and high KOSPI2 concentration in memory stocks should keep index volatility tied to AI / memory-stock volatility**. A faster leverage rebuild remains the risk to this view, but tighter regulatory scrutiny in Korea and Hong Kong should slow leverage growth and reduce the risk of another product-driven volatility amplification cycle.
- Monetizing Rich KOSPI2 Vol in a More Contained Regime**: Korea's deleveraging has shifted the market from a one-way momentum squeeze toward a more two-way consolidation phase. **With KOSPI2 implied volatility still rich and the term structure inverted, window KO calls offer lower-premium upside exposure** by monetizing near-dated barrier optionality. For neutral investors, **80%-70% / 120%-130% iron condors monetize elevated volatility through range-bound carry**, with tail risk defined by the wings.
- Dealers Stay Short Gamma in Japan; Long in Hong Kong**: NKY short-gamma **overhang** has eased, but at roughly US\$1.3bn, it remains meaningful, put-driven and downside-sensitive. In HSCEI, **dealers are long roughly US \$357mn of gamma**, helped by investor call selling.
- Korea Structured Product Issuance Rebounds in July; Japan Still Muted**: **Korea ELS issuance climbed to US\$2.2bn in July**, the highest since November 2023, led by KOSPI2-linked structures, likely helped by higher volatility and lower entry levels improving product terms. Japan issuance remained constrained at US\$37mn.

Global Equity Derivatives Strategy

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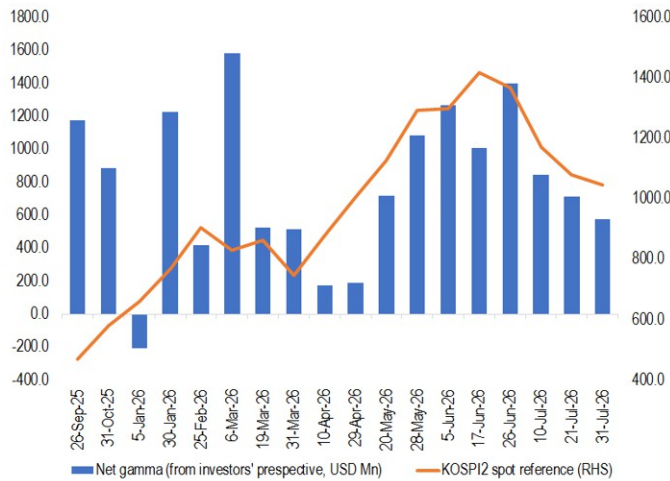
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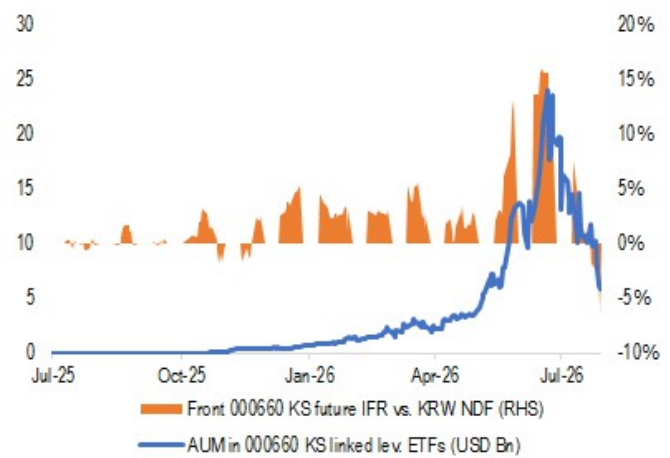
Figure 1: Korea short gamma has compressed sharply, but remains material



A positive number indicates investors (dealers) are long (short) gamma
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 2: SK hynix front-futures funding turns negative as ETF AUM unwinds

Average 10-day volume-weighted front futures-implied funding rate versus KRW NDF rate (annualized) excluding the roll period - defined as the 10 days before expiry



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

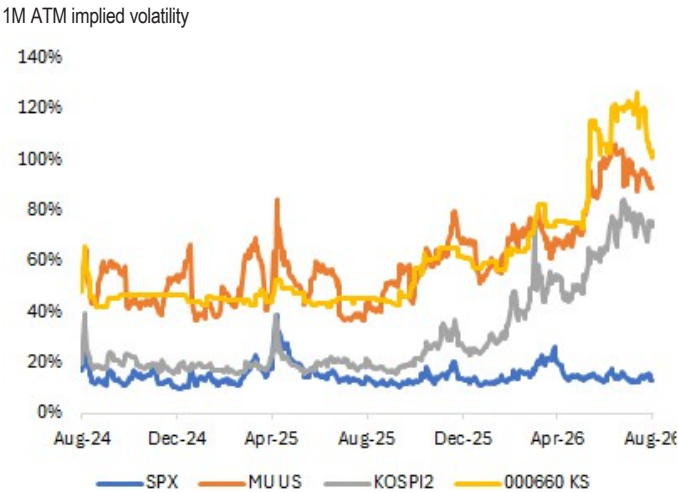
Table 1: Leveraged ETFs still dominate Korea's gamma stack; options partly offset

Estimated Korea equity-linked derivatives net gamma outstanding overview

Spot as % of current spot	60%	70%	80%	90%	100%	110%	120%	130%	140%
KOSPI2 spot	628	733	837	942	1,047	1,151	1,256	1,361	1,466
KOSPI2 options gamma (USD mn)	-32	-104	-83	-35	-12	-29	59	109	111
-Calls gamma (USD mn)	9	6	48	31	-3	-32	48	87	82
-Puts gamma (USD mn)	-42	-110	-131	-66	-9	4	12	22	29
000660 KS spot	1030800	1202600	1374400	1546200	1718000	1889800	2061600	2233400	2405200
000660 KS options gamma (USD mn)	-21	-25	-21	-15	-10	-7	-6	-5	-4
-Calls gamma (USD mn)	-21	-24	-20	-14	-8	-5	-4	-3	-2
-Puts gamma (USD mn)	0	0	-1	-1	-2	-2	-2	-2	-2
SKHY US spot	86.238	100.611	114.984	129.357	143.73	158.103	172.476	186.849	201.222
SKHY US options gamma (USD mn)	-27	-36	-42	-42	-39	-35	-32	-29	-27
-Calls gamma (USD mn)	-5	-8	-12	-16	-20	-22	-23	-23	-23
-Puts gamma (USD mn)	-22	-29	-30	-26	-19	-13	-9	-6	-4
Leveraged ETF gamma (USD mn)	211	317	422	527	633	738	844	949	1,054
-Stocks linked ETF gamma (USD mn)	69	121	173	225	276	328	380	432	483
-Index / sector linked ETF gamma (USD mn)	142	195	249	303	356	410	464	517	571
Total gamma (USD mn)	131	152	277	436	572	667	866	1,024	1,135
Gamma as % of KOSPI2 futures t/o (3MMA)	0%	0%	1%	1%	2%	2%	2%	3%	3%

A positive number indicates investors (dealers) are long (short) gamma. Data as of 31 July 2026.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 3: AI concentration keeps KOSPI2 volatility tied to memory-stock volatility



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 4: Korea lacks the scale of US option-overwrite supply
AUM in USD Bn



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

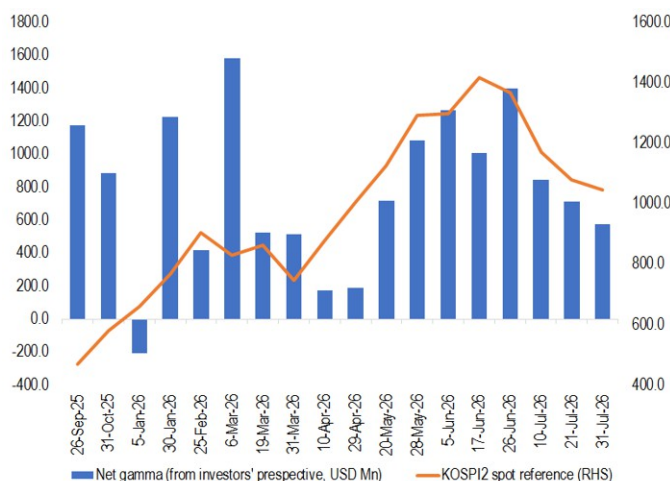
Korea Short Gamma Resets, But Remains Meaningful

Korea's leverage story has gone through a full cycle in a matter of weeks.

Leveraged ETFs built record AUM into the June peak (see [here](#)), turning the rally into a reflexive spot-up / leverage-up / volatility-up regime driven partly by rebalancing flows rather than optimism alone. As spot reversed, the same structures amplified the downside and accelerated the AUM reset. **The market now looks structurally different from late June: leverage is lower, rebalancing demand is smaller, and the gamma imbalance has compressed, but remains material.**

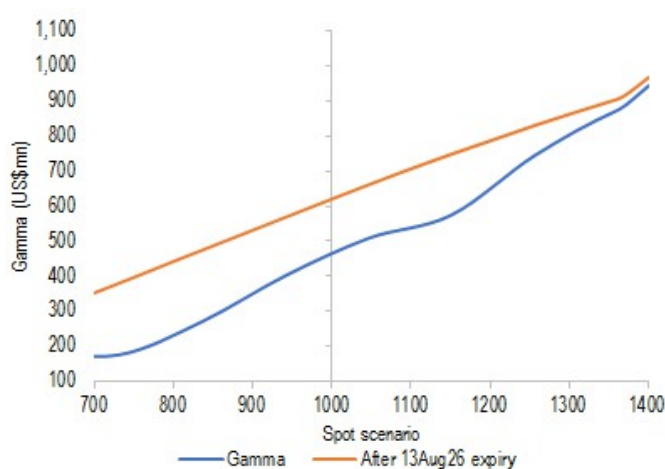
At KOSPI2 spot around 950, we estimate **Korea equity-linked dealer short gamma at roughly US\$440mn, with the short-gamma stack still led by around US\$530mn of ETF-linked dealer short gamma.** While this is well below the June peak, it remains meaningfully above the end-April lows: KOSPI2-linked dealer short gamma is around US\$260mn versus US\$105mn then, Samsung Electronics is around US\$60mn versus US\$20mn, and SK hynix is around US\$120mn versus US\$70mn.

Figure 5: Korea short gamma has compressed sharply, but remains material



A positive number indicates investors (dealers) are long (short) gamma
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 6: Short gamma is smaller, but can rebuild on rallies



A positive number indicates investors (dealers) are long (short) gamma. Data as of 31 July 2026.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

The positioning mix is also turning healthier. **Korea options have shifted toward dealer neutral-to-long gamma as investors likely monetize elevated volatility, with the long-gamma contribution most visible in SK hynix options.** We have also added Korea-linked ADR options to our gamma framework as offshore hedging channels become more relevant, with current ADR option open interest around US\$10.9bn. **The options contribution remains smaller than ETF-driven short gamma,** leaving dealers net short overall, but the direction of travel is more stabilizing than in late June.

The spot/gamma profile is largely unchanged: **ETF AUM could rebuild disproportionately on rallies,** so short gamma can rise faster on the upside than it falls on the downside. **Regulatory constraints in Asia** — including Korea's tighter rules on single-stock leveraged ETFs and Hong Kong's flexible leverage framework for CSOP

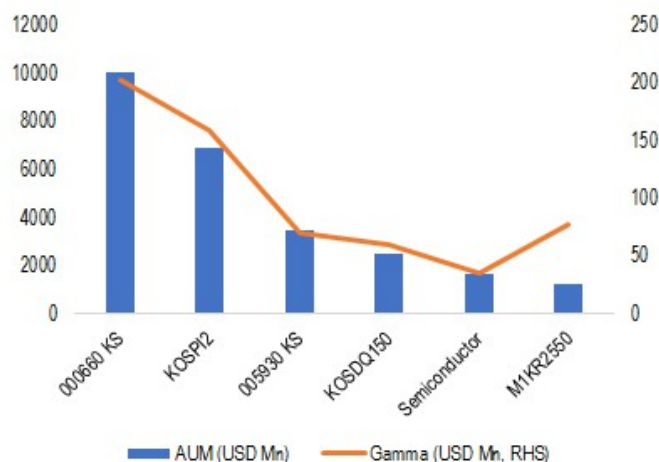
products — **could slow the rebuild of Asia-listed leverage**. However, **new US-listed leveraged ETFs highlight a developing offshore channel for Korea / AI-linked AUM** and related gamma flows, with new US-listed SK hynix-linked leveraged products gathering around US\$50mn since mid-July despite the broader AUM reset.

Figure 7: Leveraged ETF AUM reset drove the gamma compression



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 8: Korea leveraged ETF gamma by major underlying



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy. Data as of 31 July 2026.

Table 2: Leveraged ETFs still dominate Korea's gamma stack; options partly offset

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Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Table 3: Potential delta-hedging flows remain material in large spot moves

Estimated delta hedging flows under current gamma profile and spot level, by % spot move

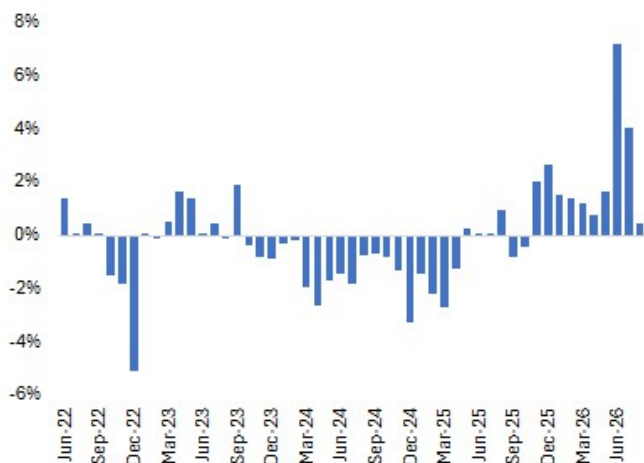
Spot change	-25%	-20%	-15%	-10%	-5%	5%	10%	15%	20%	25%
Index / sector linked (options + leveraged ETF)										
Estimated delta hedging flows (USD Bn)	-8.6	-6.9	-5.2	-3.4	-1.7	1.7	3.4	5.2	6.9	8.6
As % of KOSPI2 futures t/o (3MMA)	-24%	-20%	-15%	-10%	-5%	5%	10%	15%	20%	24%
Single stocks linked (options+leveraged ETF)										
Estimated delta hedging flows (USD Bn)	-6.0	-4.8	-3.6	-2.4	-1.2	1.2	2.4	3.6	4.8	6.0
As % of stocks cash + futures t/o (3MMA)	-12%	-10%	-7%	-5%	-2%	2%	5%	7%	10%	12%

A positive number indicates investors (dealers) are long (short) gamma. Data as of 31 July 2026
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Implied funding is another pressure gauge of the leverage cycle. The earlier leverage build pushed financing costs higher, but with leveraged ETF AUM now much lower, **long futures demand appears to have eased**. This is clearest in **SK hynix: roll-implied funding versus KRW NDF has trended lower** through the July/August and August/September rolls, and **front-futures implied funding has even turned negative** alongside the leveraged ETF AUM reset. **Index financing remains more elevated**, with KOSPI2 front futures embedding a higher funding premium and medium-tenor index funding still above normal, pointing to residual long demand and balance-sheet capacity constraints (see [here](#)).

Figure 9: SK hynix roll-implied funding has eased with the AUM reset*

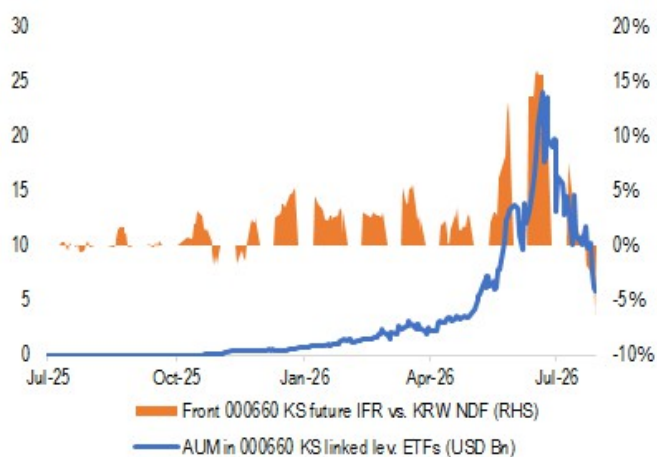
Average volume-weighted monthly futures roll-implied funding rate versus KRW NDF



*The Aug/Sep roll period is still ongoing; Aug/Sep roll data are as of 3 August 2026.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 10: SK hynix front-futures funding turns negative as ETF AUM unwinds

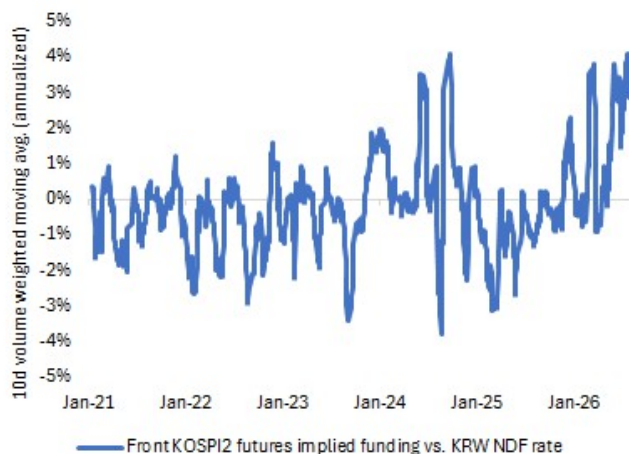
Average 10-day volume-weighted front futures-implied funding rate versus KRW NDF rate (annualized) excluding the roll period - defined as the 10 days before expiry



Data as of 30 July 2026.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 11: KOSPI2 futures continue to embed a higher funding premium

Average 10-day volume-weighted front futures-implied funding rate versus KRW NDF rate (annualized) excluding the roll period - defined as the 10 days before expiry

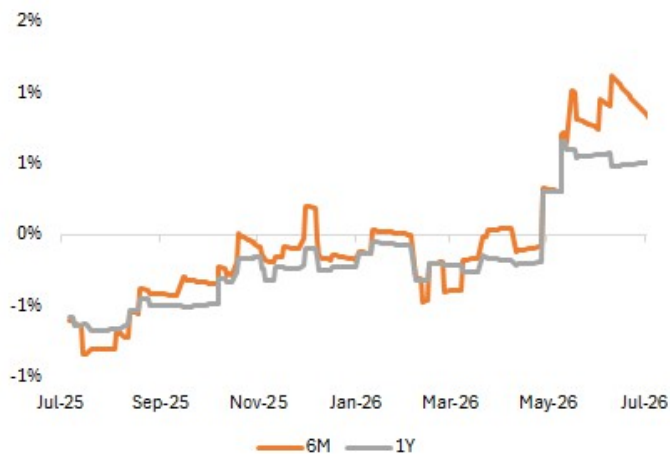


Data as of 30 July 2026

Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 12: KOSPI2 index funding remains elevated across tenors

KOSPI2 swap basis vs KRW NDF rate

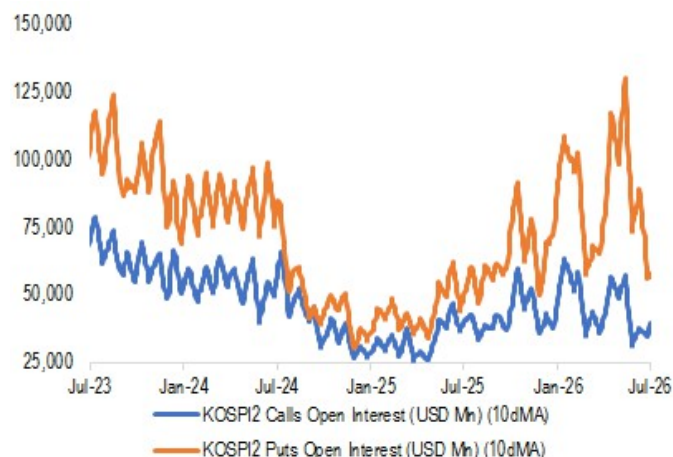


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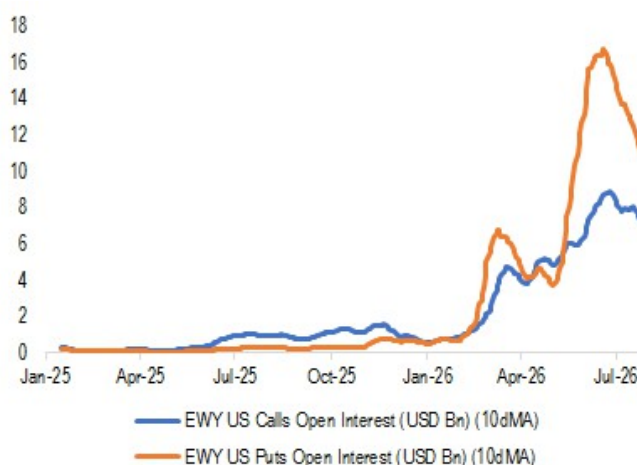
Options market activity also looks less crowded after the deleveraging. Open interest has declined from recent peaks in both onshore KOSPI2 and offshore EWY options, and recent net option selling has helped offset the ETF-driven short-gamma imbalance at the margin rather than amplify it. Investor flow patterns remain familiar: foreigners are still the key marginal buyers of KOSPI2 options, while domestic institutions remain the key sellers.

Figure 13: KOSPI2 options OI cools from the peak



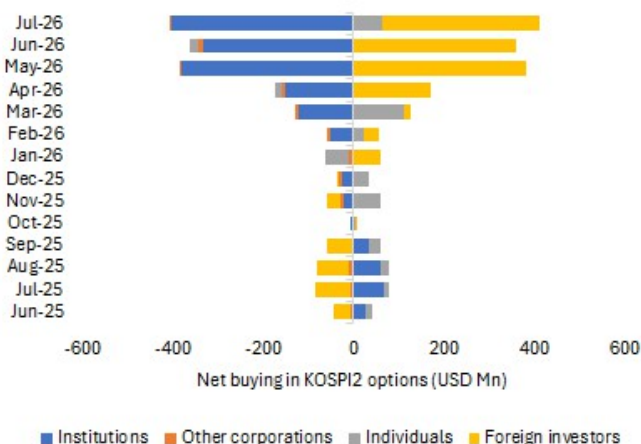
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 14: EWY options OI falls, but remains elevated



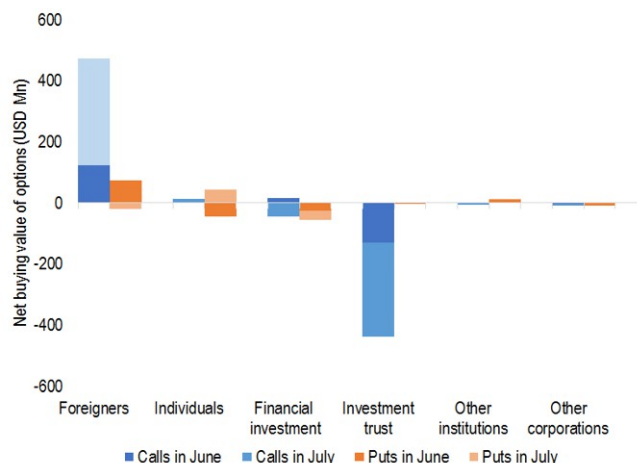
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 15: Foreigners remain the marginal buyers in KOSPI2 options



Data as of July 2026 end.
Source: KRX, Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 16: Foreign option buying remains call-led

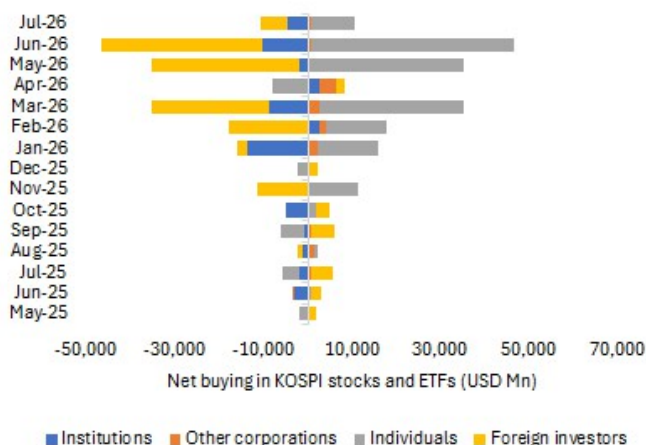


Data as of July 2026 end
Source: KRX, Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

On the cash side, monthly flows show individuals buying the dip and absorbing foreign selling, although local buying support appears to have cooled. Flows were volatile through the month, with the July 31 session seeing significant foreign buying that partly reversed earlier outflows, but foreigners remained net sellers overall.

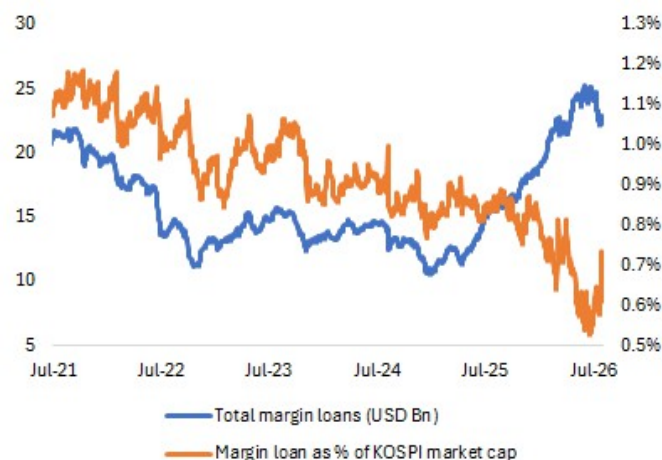
Cash-side leverage has reset too, though it remains meaningful versus history and market cap. **Margin loan balances have fallen from the peak**, echoing the broader deleveraging seen in ETF AUM and gamma. As with leveraged ETFs, part of the decline likely reflects **forced deleveraging during the July swings**, with reports citing over 1.2mn retail margin calls and more than 300k fully liquidated accounts (see [here](#))

Figure 17: Korea cash buying remains driven by individuals



Data as of July 2026 end.
Source: KRX, Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 18: Margin loans fall from the peak, but remain elevated vs market cap



Data as of July 2026 end.
Source: KRX, Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Korean investors' US equity buying has resumed after a brief slowdown, suggesting limited evidence of sustained repatriation from overseas markets back into Korea and

reinforcing the view that local buying support is becoming less intense.

Figure 19: Korean investors' US buying rebounds, limiting the Korea repatriation story



Data as of July 2026 end.
Source: KSD, J.P. Morgan Equity Derivatives Strategy.

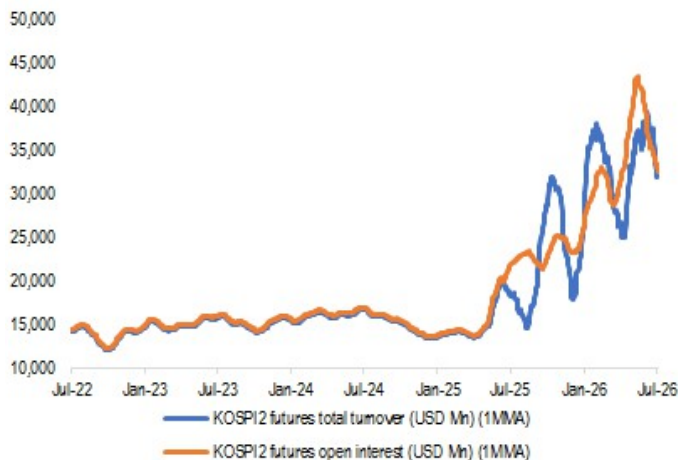
Figure 20: Annual net buying by Korean investors in overseas markets, by destination (USD Mn)

	2021	2022	2023	2024	2025	2026*
Europe	83	206	156	19	31	44
US	20792	12054	-2826	10545	32463	14510
Japan	334	-24	633	229	-1349	-313
HK/China	1039	-237	-174	-573	212	-697
China	940	-9	-58	-118	-44	-165
Other	-386	-108	-305	-118	-418	104
Total	21862	11890	-2516	10102	30939	13649

Data as of July 2026 end.
Source: KSD, J.P. Morgan Equity Derivatives Strategy.

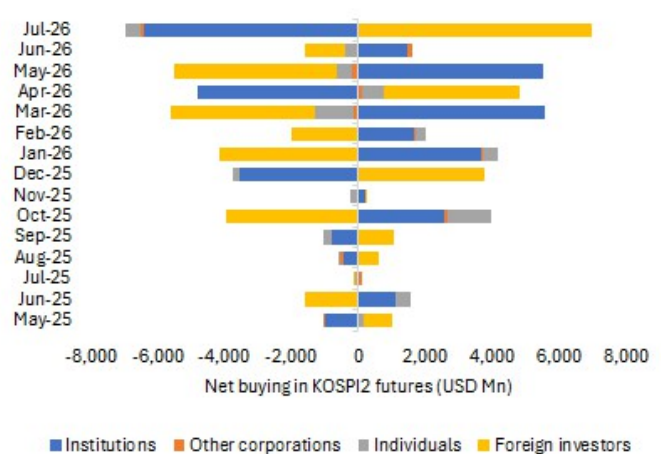
KOSPI2 futures activity has also cooled from recent highs, with both turnover and open interest easing after the July correction but remaining elevated versus pre-rally levels. Positioning has become more two-sided: **foreigners turned net buyers of KOSPI2 futures in July**, while institutions were the main sellers, reversing parts of the earlier rally pattern.

Figure 21: Turnover and open interest in KOSPI2 futures



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 22: Monthly net buying in KOSPI2 futures by investor groups



Data as of July 2026 end.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Korea Volatility: Less Explosive, Still Sticky

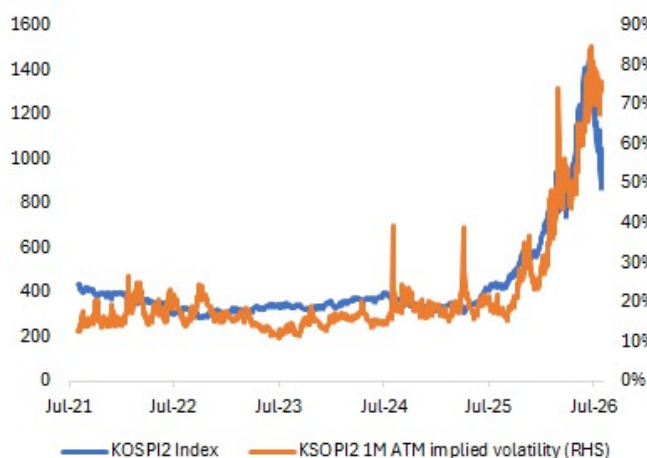
Recent deleveraging across Korea's leveraged ETF ecosystem **lowers the probability of another parabolic spot-up / volatility-up squeeze**, but it does not fully normalize Korea's volatility regime.

The volatility supply/demand balance is improving, but not restored. Leveraged ETF AUM has fallen and rebalancing demand is smaller, while structured-product issuance and early investor option selling suggest supply is beginning to return. Still, this is only one data point: supply likely needs to build more meaningfully before it can put sustained pressure on implied volatility.

Korea's volatility feedback loop was never just a leveraged ETF story. We see **four forces behind the regime: leverage, momentum, AI concentration and limited volatility supply**. The first two have eased: **leverage has reset lower** across ETF AUM, gamma and margin balances, while **momentum has cooled** after the July correction. The other two drivers — **AI concentration and limited volatility supply** — **are more structural, and help explain why Korea volatility may take time to settle** rather than quickly mean-revert to its pre-rally regime.

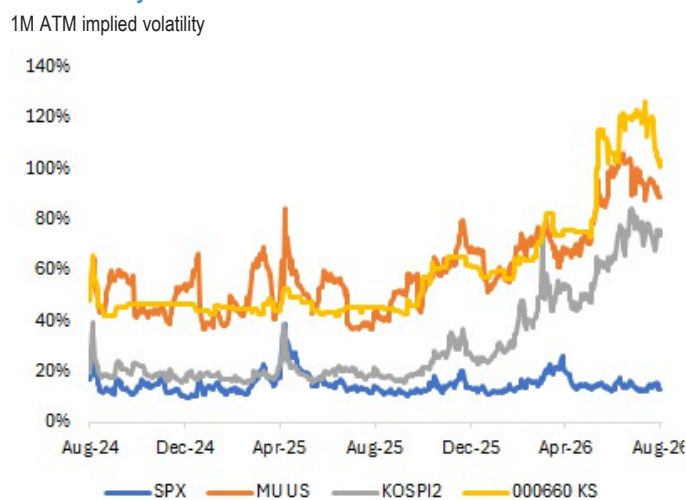
The first source of stickiness is AI concentration. **AI-linked single-stock volatility is elevated globally**, but the pass-through to index volatility is much stronger in Korea. In the US, a memory stock such as Micron has limited index volatility impact because it is less than 2% of the SPX. KOSPI2 is different: Samsung Electronics and SK hynix account for roughly 33% and 26% of the index, respectively, leaving **nearly 60% of KOSPI2 tied to the two memory leaders**. As a result, **high semiconductor volatility feeds into Korea index volatility much more directly**.

Figure 23: KOSPI2 volatility has fallen from the peak, but remains elevated



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 24: AI concentration keeps KOSPI2 volatility tied to memory-stock volatility



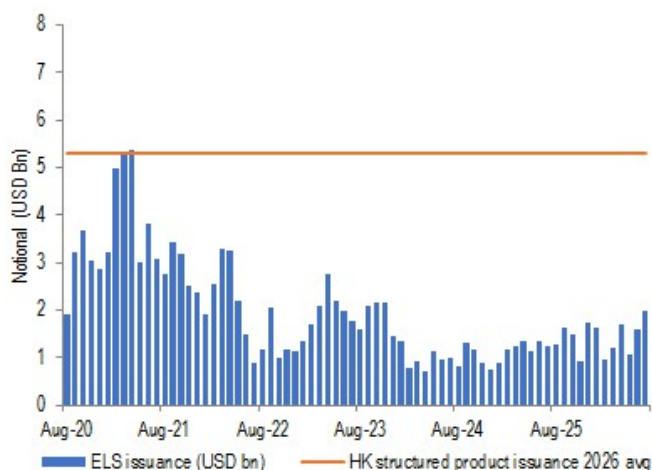
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

The second structural floor is limited volatility supply. **Korea has less of a volatility supply cushion than markets such as Hong Kong and the US**, where structured

products, overwrite strategies and systematic option selling can help volatility mean-revert more quickly after spikes. **Korea structured-product issuance has improved, but it is still running at roughly US\$1.5bn of average monthly issuance this year versus around US\$5.3bn in Hong Kong**, and remains far below prior-cycle Korea levels. **Korea also lacks the depth of the US option-overwriting ecosystem** (see [here](#)): US option-based / overwrite ETFs have grown to roughly US\$236bn of AUM across around 80 products, while Korea's covered-call ETF market is only around US\$8bn across roughly 12 ETFs.

The recent rebound in structured-product issuance and early option selling are therefore important developments to monitor. If investors continue to monetize elevated volatility more meaningfully, it could add to volatility supply and help restore part of the supply/demand imbalance. The same is true for the rebound in structured-product issuance, which is showing some signs of recovery. But for now, the supply rebuild still looks early. Until option selling and structured-product supply become more sustained, Korea volatility is likely to remain stickier than in markets with deeper volatility supply.

Figure 25: Korea ELS supply remains thin versus Hong Kong and prior cycles



Source: KSD, Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 26: Korea lacks the scale of US option-overwrite supply
AUM in USD Bn



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

The result is a regime shift, not a full normalization: **Korea is moving from elevated and explosive volatility to elevated but more contained volatility**. Deleveraging and weaker momentum should bring volatility lower, but AI concentration and limited supply mean the move lower is likely to be gradual rather than a quick mean reversion.

Recent spot/volatility behavior supports this shift toward more contained volatility. Up-beta has moderated as the leverage loop has cooled, while **July price action shows a flatter relationship between spot moves and fixed-strike volatility changes than earlier in the year**. This suggests upside moves are no longer translating into the same automatic volatility bid seen during the June squeeze.

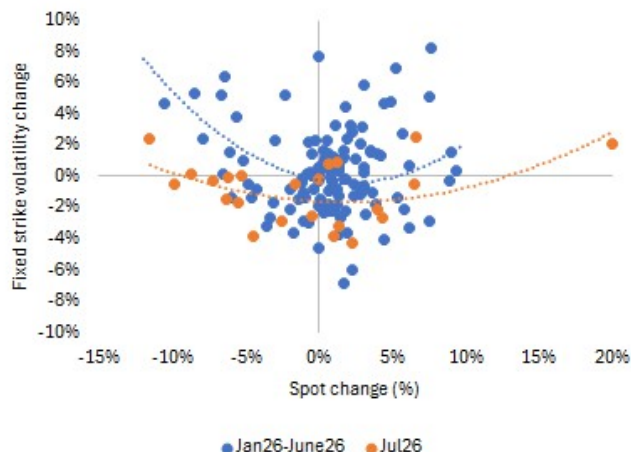
Figure 27: Spot/volatility beta has moderated as the leverage loop cools



Up-beta and down-beta are calculated using a 3M rolling window of daily changes in 1M fixed-strike volatility versus daily KOSPI2 spot returns, using only spot-up days for up-beta and spot-down days for down-beta.

Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 28: July price action shows a less reflexive spot/volatility relationship



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

The **main risk to this view is a rapid rebuild in leverage, but regulatory scrutiny should slow the next cycle**. Korea has halted new leveraged ETF listings, raised minimum deposit requirements, and pushed ETF managers to spread rebalancing through the day rather than concentrate flows near the close (see [here](#), [here](#), [here](#)). Proposed measures could go further, with caps on retail exposure to single-stock leveraged ETFs at 20% of financial assets (see [here](#)) and an “emergency action authority” that would allow regulators to temporarily cut leverage from 2x to 1.5x, and potentially as low as 1.1x, during periods of heightened market volatility (see [here](#)).

Regionally, **Hong Kong is moving in a similar direction**: following SFC guidance, CSOP will **adopt a flexible leverage framework** from August 3 across 12 leveraged / inverse ETFs, including Samsung and SK hynix-linked products, allowing **leverage to be adjusted daily from a maximum of 2x to as low as 1.1x in extreme markets** (see [here](#)). The aim is to reduce tracking-difference risk during turbulent trading, reinforcing the broader regulatory push to limit product-level amplification when volatility is high.

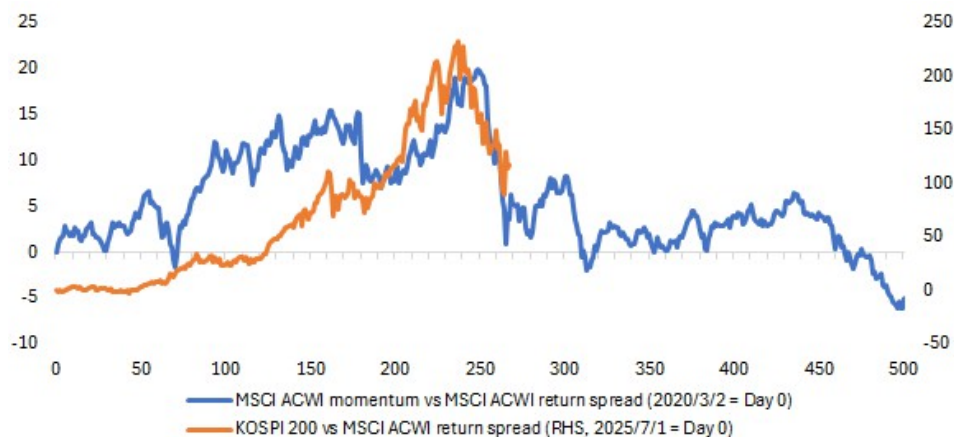
Monetizing Rich KOSPI2 Vol in a More Contained Regime

Following one of the strongest year-to-date rallies globally, Korean equities have entered a period of deleveraging as crowded positioning unwinds. While near-term **volatility is likely to remain elevated**, we believe the market is transitioning from an explosive momentum-driven phase to a more contained regime characterized by **consolidation and two-way trading**.

A useful historical analogue is the **1Q21 global momentum correction**, when a sharp rise in US Treasury yields triggered a sell-off in high-growth stocks. Following the initial drawdown, momentum experienced a **partial rebound** before **undergoing a second correction** and **subsequently traded sideways for several months**. Although the current episode differs in both catalyst and magnitude, it highlights a common market pattern: after the first relief rally, **markets often consolidate before establishing the next directional trend**. This continues our recent theme of fading extreme Korea volatility moves and positioning for a further decline in volatility as the leverage-volatility feedback loop becomes less explosive (see [here](#)).

With KOSPI2 implied volatility still screening as exceptionally rich, we favor option **structures that monetize elevated volatility** while maintaining exposure to a gradual medium-term recovery.

Figure 29: In 1Q21, momentum sold off, rebounded, then retreated again before months of sideways trading—a consolidation pattern Korean equities may now be echoing



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

Window KO Calls for efficient upside

Window knock-out calls offer an efficient way to retain bullish KOSPI2 exposure while monetizing still-rich volatility. The embedded KO feature monetizes elevated implied volatility and reduces cost versus vanilla calls, while the limited observation window helps avoid paying for a full-life barrier. The structure is especially attractive in the current setup because the inverted term structure makes near-dated barrier optionality valuable to sell. The proposed structure **offers roughly 60% premium saving versus the comparable vanilla call**.

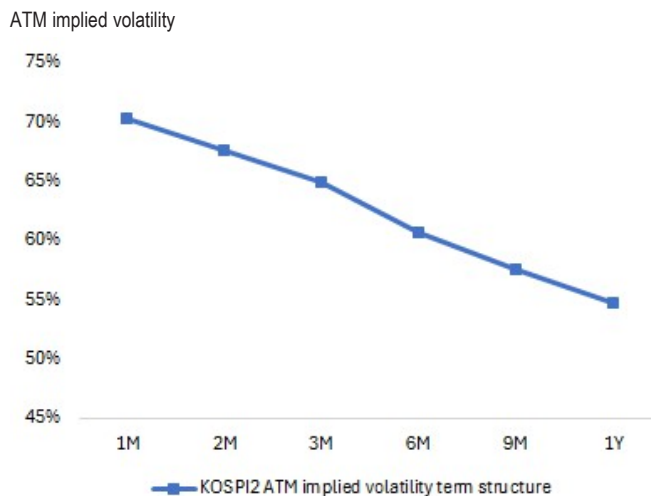
- Buy 08Oct26 KOSPI2 110% call with window KO at 130%, continuously observed from trade date to 10Sep26 @3.3% offer

Figure 30: KOSPI2 spot with proposed window KO call strikes and barrier



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

Figure 31: Inverted KOSPI2 volatility term structure supports window KO calls



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

Iron Condor for Range-Bound Volatility Carry

For investors with a more neutral view, we like iron condors as a way **to monetize still-rich KOSPI2 volatility while defining tail risk**. The iron condor is designed for a wider range-bound outcome and benefits from time decay and volatility normalization as long as spot stays within the short strikes.

The proposed structure collects 2.3% premium by selling the 80%-120% range and buying wings at 70% and 130%. The profit is maximized if KOSPI2 finishes between 80% and 120% of initial spot. Breakevens are around 77.7% and 122.3% after premium, with maximum loss capped at around 7.7% on either side.

- Sell 10Sep26 KOSPI2 80%-70% put spread and 120%-130% call spread @2.3% bid (71.6v/77.3v/64.3v/66.3v)

Dealers Stay Short Gamma in Japan; Long in Hong Kong

NKY Short Gamma Eases After the Pullback

Japan's short-gamma overhang eased in July as spot pulled back from the June highs. Around NKY spot 64,000, we now estimate dealers hold roughly **US\$1.3bn of net short gamma**, equivalent to approximately 5% of futures turnover. The imbalance is lower than in June, **but remains meaningful** and is still driven mainly by put options, while ETF-driven gamma remains muted in Japan.

The current **gamma profile remains more downside-sensitive than upside-sensitive**. Dealer short gamma is close to its local peak around current spot and should fall if spot rises, but can remain around these levels on a moderate decline in spot.

Figure 32: Investors' gamma positioning in Japan equities

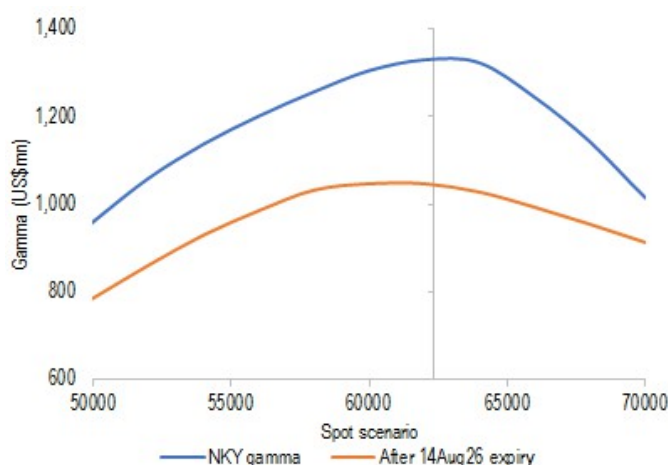
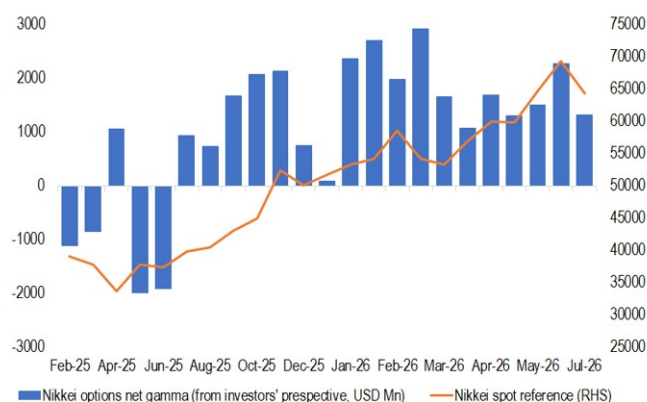


Figure 33: Investors' long (dealers' short) gamma in NKY eases with lower spot



A positive number indicates investors (dealers) are long (short) gamma. Data as of 31 July 2026.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

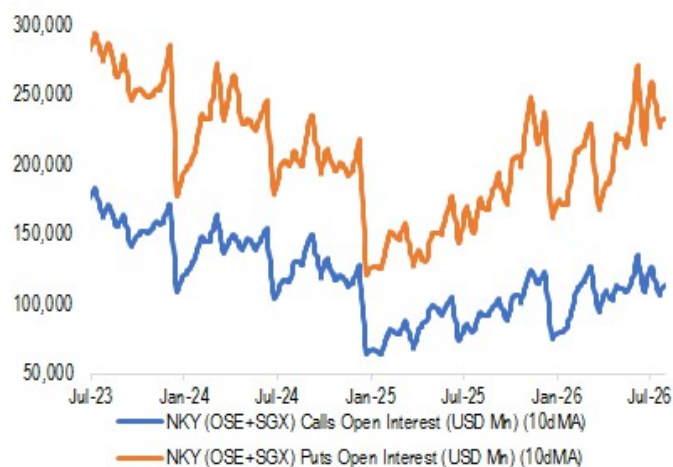
A positive number indicates investors (dealers) are long (short) gamma. Data as of 31 July 2026.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Table 4: Estimated Japan equities-linked derivatives net gamma outstanding overview

Nikkei spot	56,000	58,000	60,000	62,000	64,000	66,000	68,000	70,000	72,000
Listed options gamma (USD mn)	1,068	1,119	1,162	1,182	1,171	1,087	982	850	733
-Calls gamma (USD mn)	256	300	331	346	337	307	274	255	250
-Puts gamma (USD mn)	812	819	831	836	833	781	707	595	483
Leveraged ETF gamma (USD mn)	134	139	144	149	153	158	163	168	173
Total gamma (USD mn)	1,202	1,258	1,306	1,331	1,324	1,246	1,145	1,018	905
Gamma as % of NKY futures t/o (3MMA)	5%	5%	5%	5%	5%	5%	4%	4%	4%

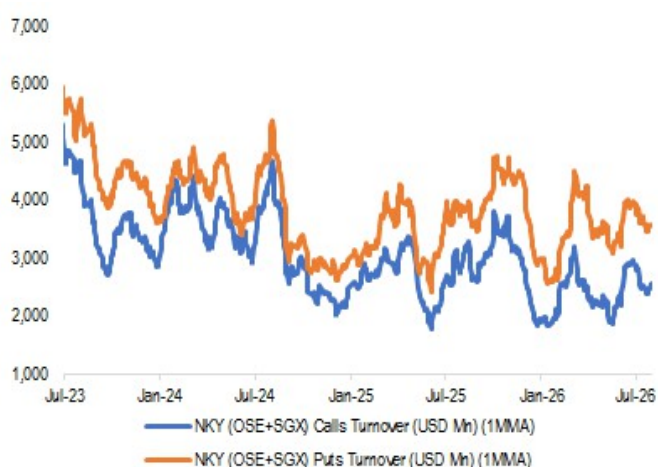
A positive number indicates investors (dealers) are long (short) gamma. Data as of 31 July 2026.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy.

Figure 34: NKY options open interest



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

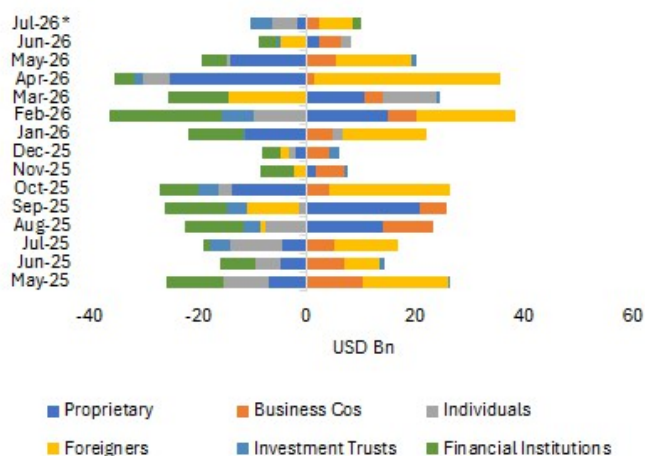
Figure 35: NKY options turnover



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

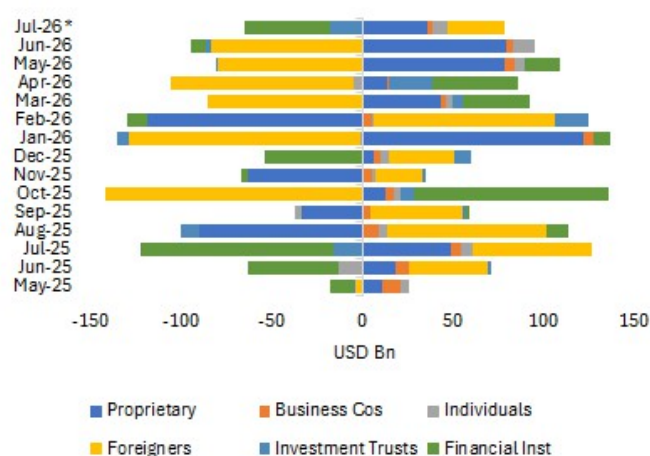
Broader flows show foreigners returning amid the selloff, possibly helped by better entry levels after the correction. Foreigners **bought both cash equities and NKY futures in July**, while individuals and investment trusts also bought cash equities into the pullback. In futures, financial institutions were the main sellers on the other side of foreign buying.

Figure 36: Foreigners resume buying Japanese cash equities



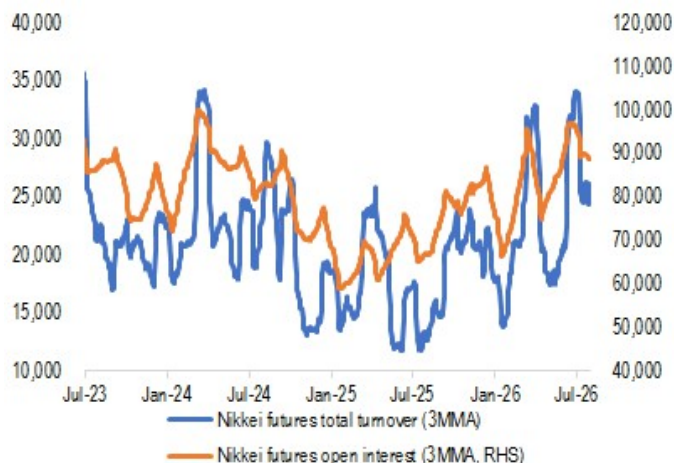
Data as of 24 July 2026.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

Figure 37: Foreigners turn buyers of NKY futures in July



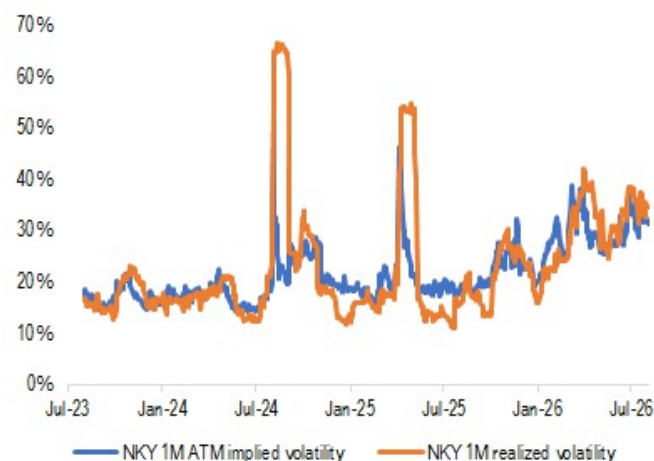
Data as of 24 July 2026.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

Figure 38: Turnover and open interest in NKY futures



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

Figure 39: NKY implied and realized volatility



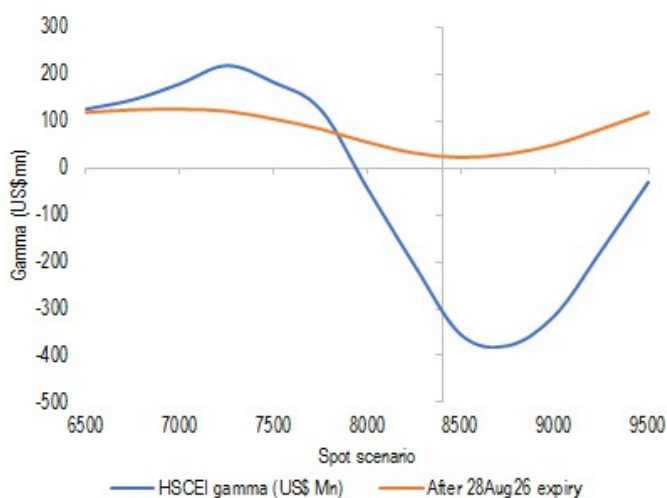
Source: Bloomberg Finance L.P., J.P. Morgan Global Markets Strategy

Dealers Turn Long Gamma as HSCEI Rallies

HSCEI was the stronger leg in July, supported by renewed interest in Hong Kong / China tech and internet stocks, rotation into lower-valuation tech names, and policy / capital-market support headlines. Options activity, however, has not revived in the same way. HSCEI options open interest and turnover remain muted, and dealers are now long gamma.

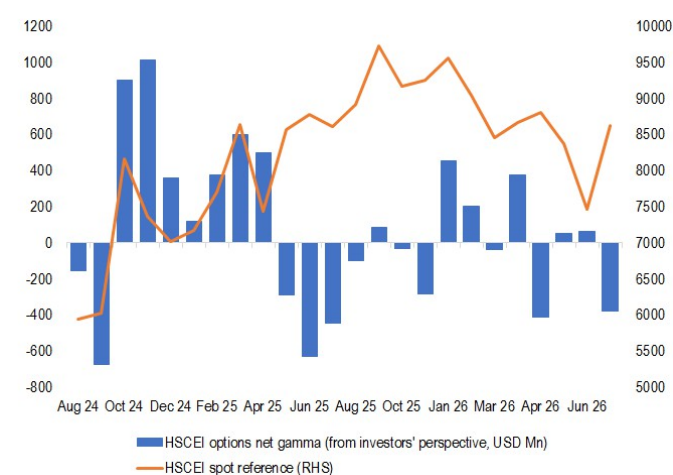
Around HSCEI spot 8,500, we estimate dealers are long roughly US\$357mn of gamma, equivalent to around 4% of futures turnover. Recent flows suggest the dealer long-gamma position is largely driven by investors net selling of call options. This positive dealer gamma should help dampen hedging flows and keep HSCEI realized volatility relatively contained.

Figure 40: Investors' gamma positioning in Hong Kong equities



A positive number indicates investors (dealers) are long (short) gamma. Data as of 31 July 2026.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

Figure 41: Investors' (dealers') gamma history in Hong Kong equities



A positive number indicates investors (dealers) are long (short) gamma. Data as of 31 July 2026.
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

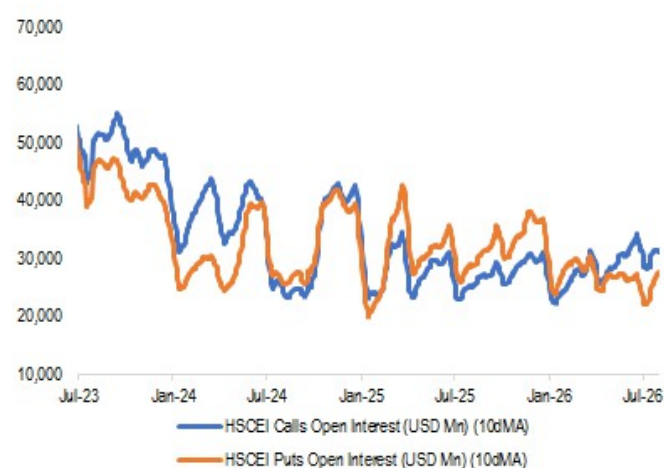
Table 5: Estimated HSCEI-linked derivatives net gamma outstanding overview

HSCEI spot	7,500	7,750	8,000	8,250	8,500	8,750	9,000	9,250	9,500
Listed options gamma (USD mn)	28	-24	-178	-327	-461	-496	-449	-327	-194
-Calls gamma (USD mn)	4	-24	-130	-238	-347	-378	-346	-249	-142
-Puts gamma (USD mn)	23	-1	-48	-89	-115	-118	-103	-78	-52
Leveraged ETF gamma (USD mn)	132	136	141	145	150	154	159	163	167
Total gamma (USD mn)	182	124	-44	-208	-357	-381	-316	-177	-31
Gamma as % of futures t/o (3MMA)	2%	1%	0%	-2%	-4%	-4%	-3%	-2%	0%

A positive number indicates investors (dealers) are long (short) gamma. Data as of 31 July 2026.

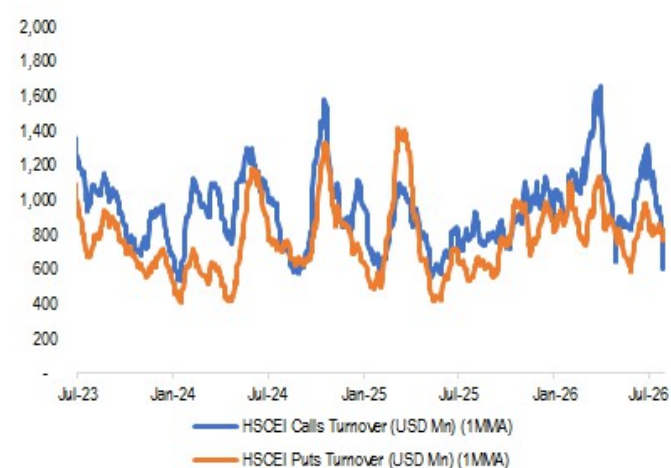
Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

Figure 42: HSCEI options open interest



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

Figure 43: HSCEI options turnover



Source: Bloomberg Finance L.P., J.P. Morgan Equity Derivatives Strategy

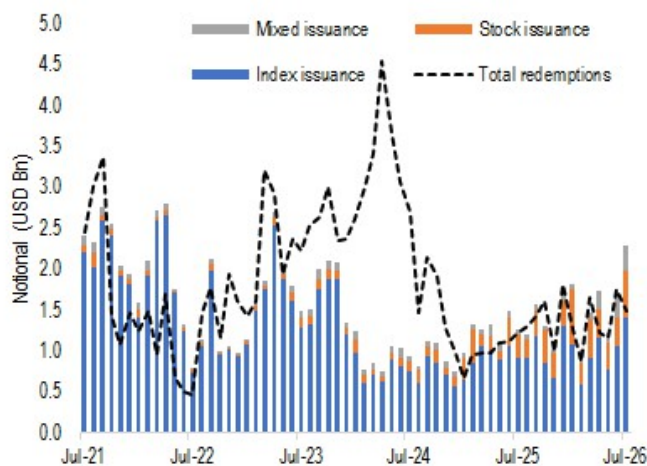
Korea Structured Product Issuance Rebounds in July; Japan Still Muted

Structured Product Issuance Trends in Korea and Japan

Korea ELS issuance rebounded sharply in July, with new issuance rising to US \$2.2bn, the highest since November 2023, and exceeding redemptions of US\$1.5bn, turning net issuance positive for the month. The pickup was led by index-linked products, which contributed around US\$1.4bn, with **KOSPI2-linked structures dominating** as the combination of higher volatility and lower entry levels after the correction likely improved product terms.

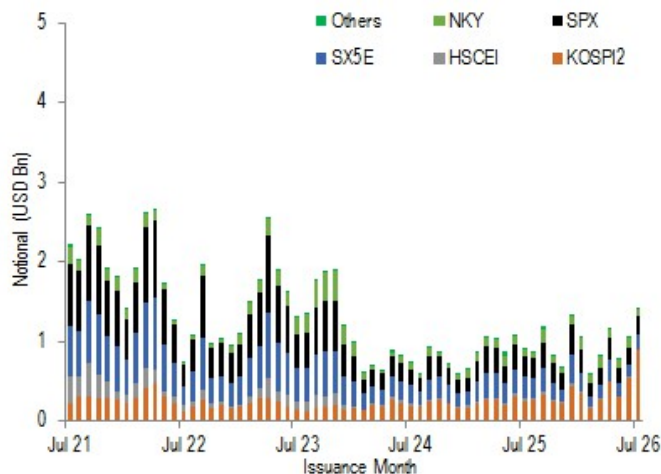
Single-stock issuance also strengthened, rising to around US\$580mn, with demand still skewed toward domestic names. **SK hynix and Samsung Electronics** saw the largest July issuance, followed by Tesla among overseas underlyings. Overall, domestic stocks accounted for US\$314mn of single-stock issuance, versus US\$269mn for overseas names.

Figure 44: Korean ELS issuance and redemption history



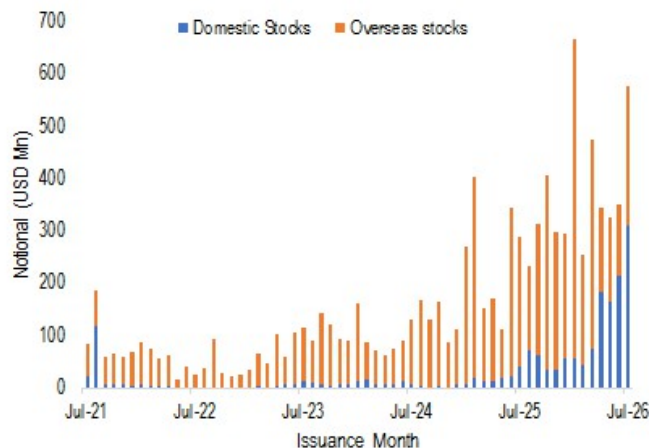
Source: J.P. Morgan Equity Derivatives Strategy, KSD. *Based on issuance data as of July 2026 end.

Figure 45: Korean ELS issuance by major index underlying



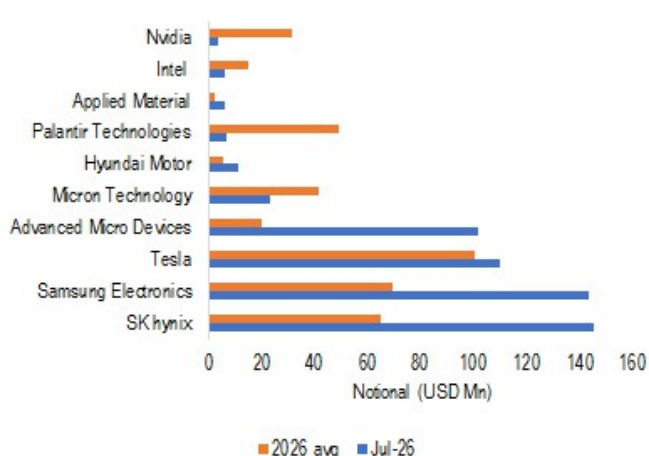
Source: J.P. Morgan Equity Derivatives Strategy, KSD. *Based on issuance data as of July 2026 end.

Figure 46: Korea ELS issuance trend for stock underlyings



Source: J.P. Morgan Equity Derivatives Strategy, KSD. *Based on issuance data as of July 2026 end.

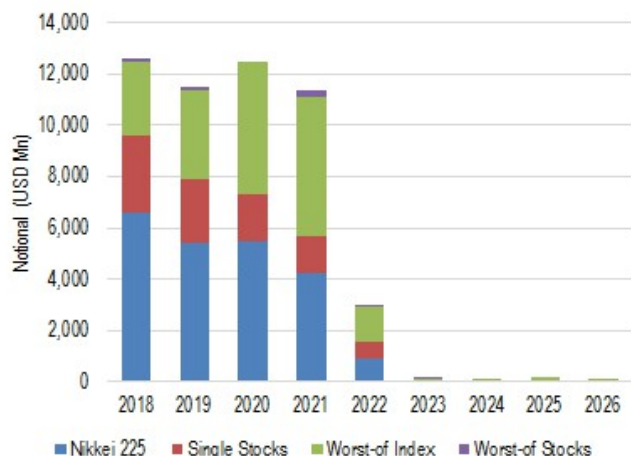
Figure 47: Top 10 stocks by issuance notional in July 2026



Source: J.P. Morgan Equity Derivatives Strategy, KSD. *Based on issuance data as of July 2026 end.

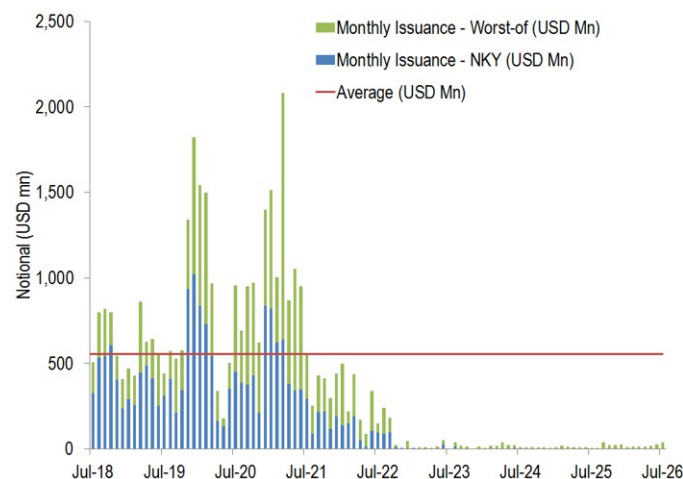
Japan: Structured product issuance remained extremely limited in July 2026, with only US\$37mn of product notional issued during the month.

Figure 48: Japan structured product issuance history



Source: J.P. Morgan Equity Derivatives Strategy. *Based on issuance data as of July 2026 end. The data are for public offerings only.

Figure 49: NKY structured product issuance by month



Source: J.P. Morgan Equity Derivatives Strategy. *Based on issuance data as of July 2026 end. The data are for public offerings only.

Risk Profile of Structured Product Indices

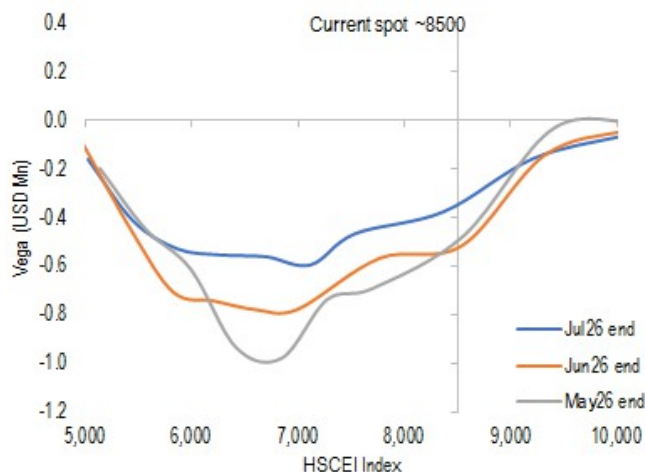
We estimate that **structured products have approximately US\$0.4mn in vega outstanding in HSCEI** at a spot level of around 8,500. Dealers' current exposure to HSCEI vega risk is relatively low, mainly due to the limited outstanding notional, as a substantial amount of products expired last year without corresponding reinvestment. Given the low outstanding notional, we anticipate minimal impact from product knock-outs.

Outstanding vega tied to KOSPI2 is estimated at around US\$14mn at spot ~950, with vega notional rising in July due to higher issuance in KOSPI2-linked

products. Given the current profile, we estimate that a 10% decline/rise in spot would generate potential vega supply/demand of about US\$2mn. On the redemption side, upcoming knock-out observations over the next three months are concentrated around 720-840, with outstanding notional of approximately US\$930mn. Whether knock-outs occur will depend on the performance of the other indices in the worst-of baskets.

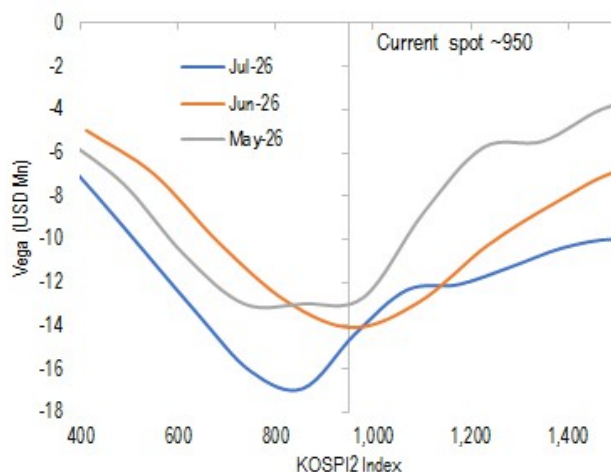
At spot around 64,000, **dealers' vega exposure in NKY-linked Korean and Japanese structured products is estimated at about US\$2.5mn.** We estimate that a 10% decrease in spot would result in potential vega supply of approximately US\$1.3mn, while a 10% increase in spot would lead to potential vega demand of about US\$0.8mn. On the redemption side, upcoming knock-out observations over the next three months are largely concentrated at barriers above 45,000, with outstanding notional around US\$866mn. Whether these knock-outs are triggered will depend on the performance of the other indices referenced in the worst-of baskets.

Figure 50: Estimated vega profile for HSCEI-linked structured products (from the perspective of products)



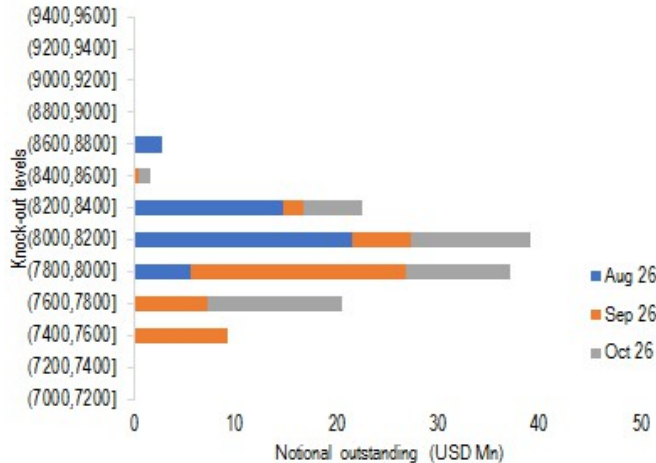
Source: KSD, J.P. Morgan Equity Derivatives Strategy. *Based on issuance data as of July 2026 end.

Figure 51: Estimated vega profile for KOSPI2-linked structured products (from the perspective of products)



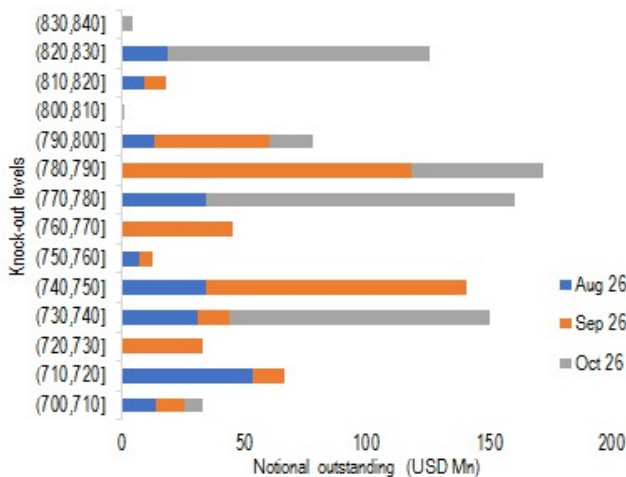
Source: KSD, J.P. Morgan Equity Derivatives Strategy. *Based on issuance data as of July 2026 end.

Figure 52: Distribution of outstanding notional for HSCEI-linked structured products by knock-out level



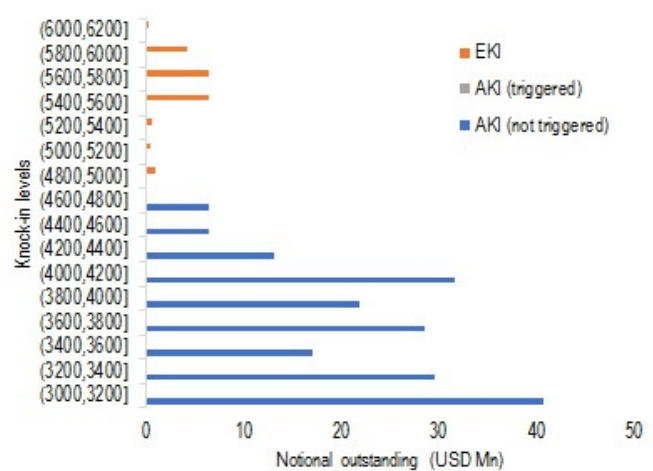
Source: KSD, J.P. Morgan Equity Derivatives Strategy. *Based on issuance data as of July 2026 end. Current spot level is at ~8500.

Figure 54: Distribution of outstanding notional for KOSPI2-linked structured products by knock-out level



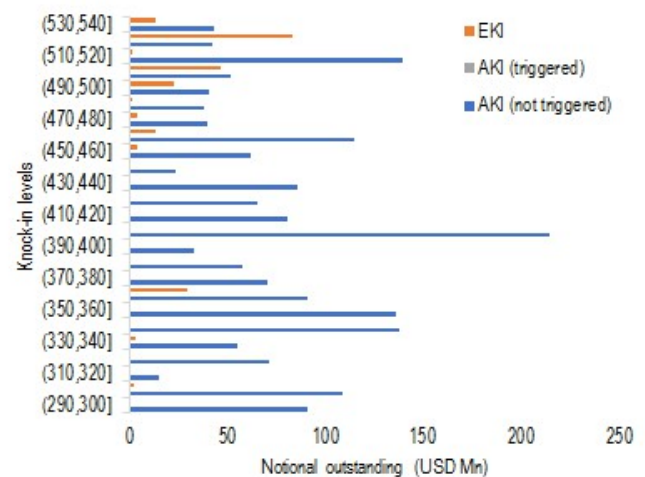
Source: KSD, J.P. Morgan Equity Derivatives Strategy. *Based on issuance data as of July 2026 end. Current spot level is at ~950.

Figure 53: Distribution of outstanding notional for HSCEI-linked structured products by knock-in level



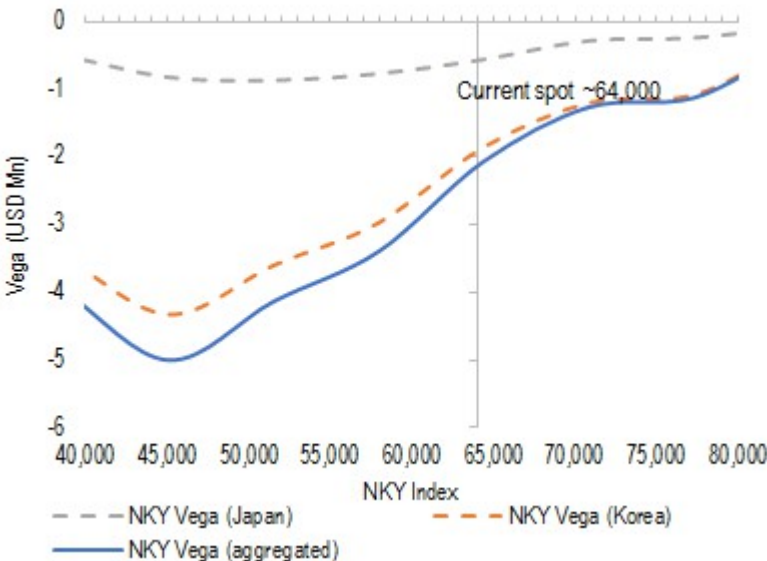
Source: KSD, J.P. Morgan Equity Derivatives Strategy. *Based on issuance data as of July 2026 end. Current spot level is at ~8500.

Figure 55: Distribution of outstanding notional for KOSPI2-linked structured products by knock-in level



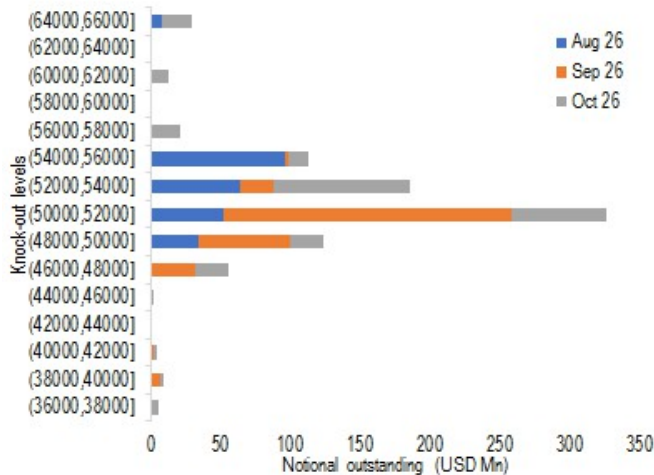
Source: KSD, J.P. Morgan Equity Derivatives Strategy. *Based on issuance data as of July 2026 end. Current spot level is at ~950.

Figure 56: Estimated vega profile for NKY-linked structured products (from the perspective of products)



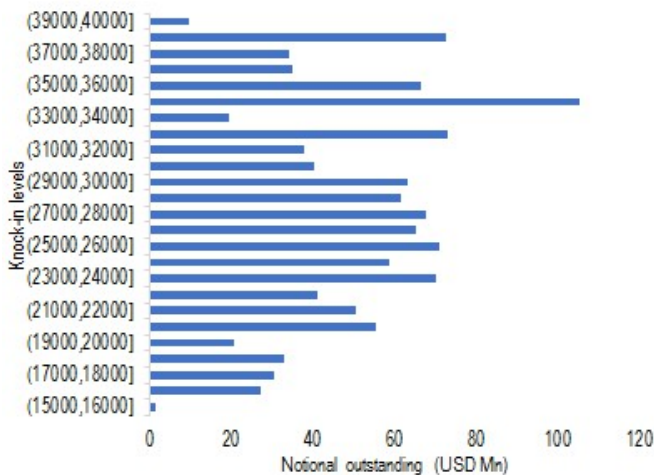
Source: KSD, J.P. Morgan Equity Derivatives Strategy. *Based on issuance data as of July 2026 end

Figure 57: Distribution of outstanding notional for NKY-linked structured products by knock-out level



Source: KSD, J.P. Morgan Equity Derivatives Strategy. *Based on issuance data as of July 2026 end. Current spot level is at ~64000.

Figure 58: Distribution of outstanding notional for NKY-linked structured products by knock-in level



Source: KSD, J.P. Morgan Equity Derivatives Strategy. *Based on issuance data as of July 2026 end. Current spot level is at ~64000.

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IB clients**	95%	92%	87%

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